Total Marks: 30 Time: 2 hours

Note: Attempt all questions and write answers within provided spaces on E-Sheet.

SECTION A (6 x 3 = 18)

Q.1 A student carried out biochemical tests on the given food material to find out its nutrient contents.

a) Re-draw table 1.1 on E-sheet and complete with required information. [3]

Table: 1.1

Reagent	Used to test	Observations
Absolute ethanol and cold water		
	Protein	
Benedict's solution		

b) A student's test on food sample gives colour change with iodine. Does it indicate anything about the presence of glucose in the sample? If your answer is no, then, which biochemical test would the student have to perform to find out about the presence of glucose? [2]

c) Identify the enzyme catalyzing the reaction shown in figure 1.1? [1]

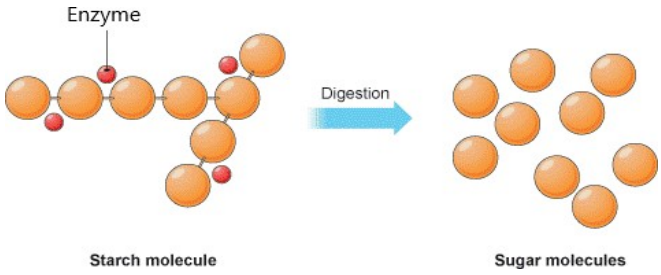


Figure: 1.1

Q2. A pot with a small plant was placed horizontally in dark condition. After few days, roots bended downwards and shoot grew upwards.

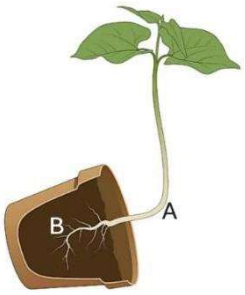


Figure: 2.1

a) Identify the plant movements shown at point A and B in figure 2.1 and define them? [2]

b) Relate these movements shown in figure 2.1with the type of stimuli. [1]

c) Give one benefit of each type of movement to the plant shown in above figure? [2]

d) Name any other type of tropic movement in plants. [1]

Q3. A student studied the effect of limiting factors on rate of photosynthesis and obtained following data shown in **Table 3.1**

Concentration of CO_2 (% age)	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14
Rate of photosynthesis in **LOW LIGHT INTENSITY**	0	20	29	35	39	42	45	47
Rate of photosynthesis in **HIGH LIGHT INTENSITY**	0	22	35	47	68	84	94	99

a) Make two graphs of above data to show rates of photosynthesis in low and high intensities of light. Compare the rates of photosynthesis in both graphs. [2]

b) Label the region of graphs that show carbon dioxide as limiting factor. [2]

c) Name two factors other than CO_2 and light intensities, that may act as limiting factor for rate of photosynthesis. How a factor becomes limiting factor for the process of photosynthesis? [2]

SECTION-B (12 Marks)

Q.4

a) Identify which type of germination is shown in figure 4.1. Identify labelled parts A, B and C. [2]

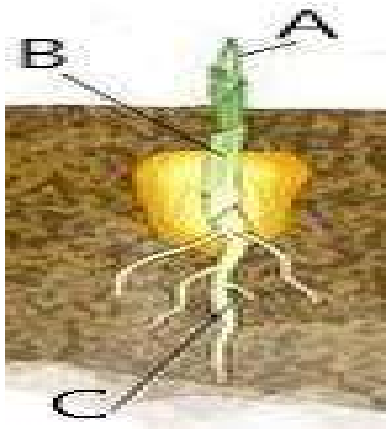


Figure: 4.1

b) Identify the organism shown in figure 4.2. [1]



Figure: 4.2

c)

d) Identify the type of plant tissue shown in figure 4.3. [1]

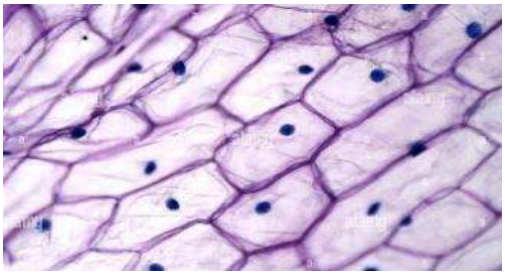


Figure: 4.3

Q.5

a) The following diagram shows diffusion of ink in two glasses of water at different temperatures. Which glass will show faster diffusion? Give a reason for your answer. [2]

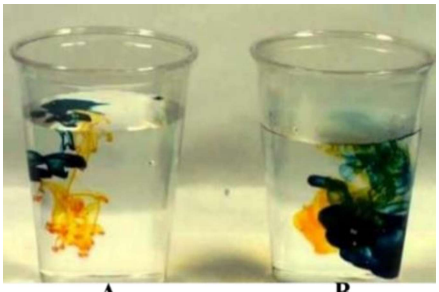


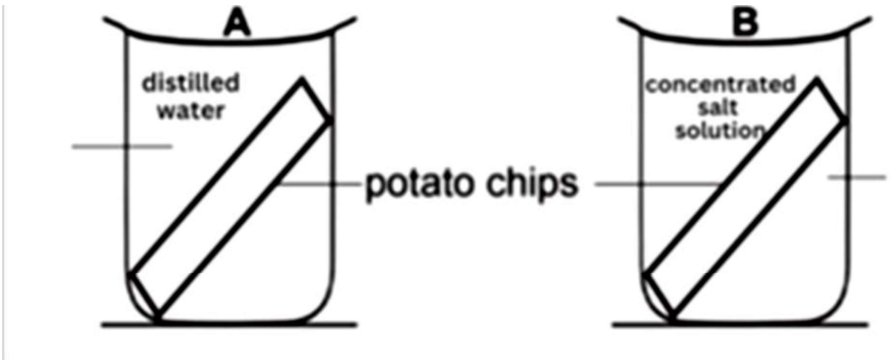
Figure: 5.1

b) While recording the height of your classmates, you obtained the data given in table 4.1 and draw a histogram using your scale to represent this data. [2]

Table: 5.1

Height range in inches	Number of students
59-60	2
61-62	5
63-64	12
65-66	18
67-68	9
69-70	3
71-72	1

Q6. (a) The figure 4.1 shows process of osmosis. Identify the glass beakers in which exosmosis and endosmosis will occur. Give reasons of your answer. [2]



. Fig 6.1

(b) Name the three parts of dicot seed labelled as A, B and C indicated in diagram 6.2. Give the role of labelled part C in seed germination.[2]

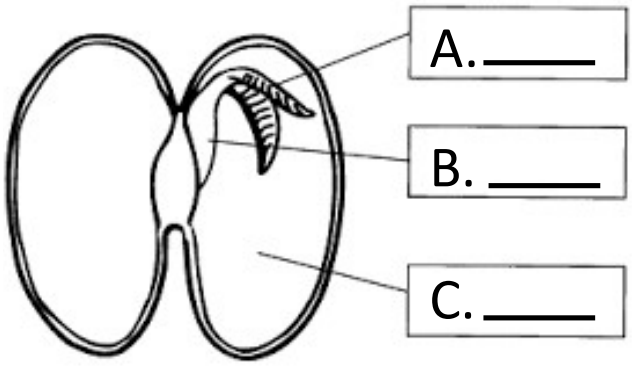


Fig:6.2

COMPLETE EXAM-STYLE SOLUTIONS • 30 MARKS

Biology SSC (Composite)
New Model Paper: Practical Based Assessment

For Annual Examination 2027 & onwards • Time: 2 hours

The printed paper specifies 2027 & onwards, although the uploaded filename contains 2026. All six questions are solved below in the original order. Section A = 18 marks; Section B = 12 marks.

Q1(a). Complete Table 1.1: biochemical food tests [3]

Reagent	Used to test	Positive observation
Absolute ethanol, then cold water	Lipids / fats	A cloudy, milky-white emulsion appears.
Biuret reagent (sodium hydroxide and dilute copper sulfate)	Protein	The blue reagent changes to lilac / violet / purple.
Benedict's solution, heated in a hot-water bath	Reducing sugars, e.g. glucose	Blue changes to green, yellow, orange or a brick-red precipitate, depending on sugar concentration.

Method detail: For the emulsion test, first shake the sample with ethanol, then add the ethanol extract to water. For Benedict's test, heating in a water bath is essential.

Q1(b). Does a positive iodine test show glucose? Which test is needed? [2]

No. A blue-black colour with iodine indicates **starch**, not glucose. It does not establish whether glucose is present or absent in the same sample.

Test a **fresh portion** of the food extract with **Benedict's solution** and heat it in a hot-water bath. A green to brick-red colour/precipitate indicates reducing sugar, such as glucose. If no reducing sugar is detected, it remains blue.

Benedict's is the expected school-level glucose test here, but it detects other reducing sugars too; it is not uniquely specific to glucose.

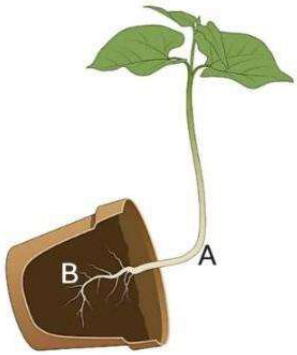
Q1(c). Identify the enzyme acting on starch in Figure 1.1 [1]

Amylase. It hydrolyses starch into smaller sugars, principally maltose and shorter carbohydrate chains. The diagram alone does not identify the enzyme's salivary or pancreatic source.

Starch + water —amylase→ **smaller sugars**. Do not write protease: protease digests protein.

Q2. Tropic movements in a horizontal plant [6]

Situation: A small potted plant is placed horizontally in the dark. Later, its shoot grows upwards and roots bend downwards.



(a) Identify and define the movements at A and B [2]

Point	Movement	Definition
A: shoot	Negative geotropism / negative gravitropism	Directional growth of the shoot away from the direction of gravitational pull.
B: root	Positive geotropism / positive gravitropism	Directional growth of the root towards the direction of gravitational pull.

(b) Relate both movements to the stimulus [1]

The stimulus is **gravity**. The root grows in the direction of gravity, while the shoot grows against it. The plant was kept in the dark, so directional light is not the cause of the bends shown.

(c) One benefit of each movement [2]

Shoot: Upward growth helps the shoot emerge above the soil and position leaves where they can receive light for photosynthesis when light becomes available.

Root: Downward growth anchors the plant and places roots in the soil for absorption of water and mineral ions.

(d) Name another tropic movement [1]

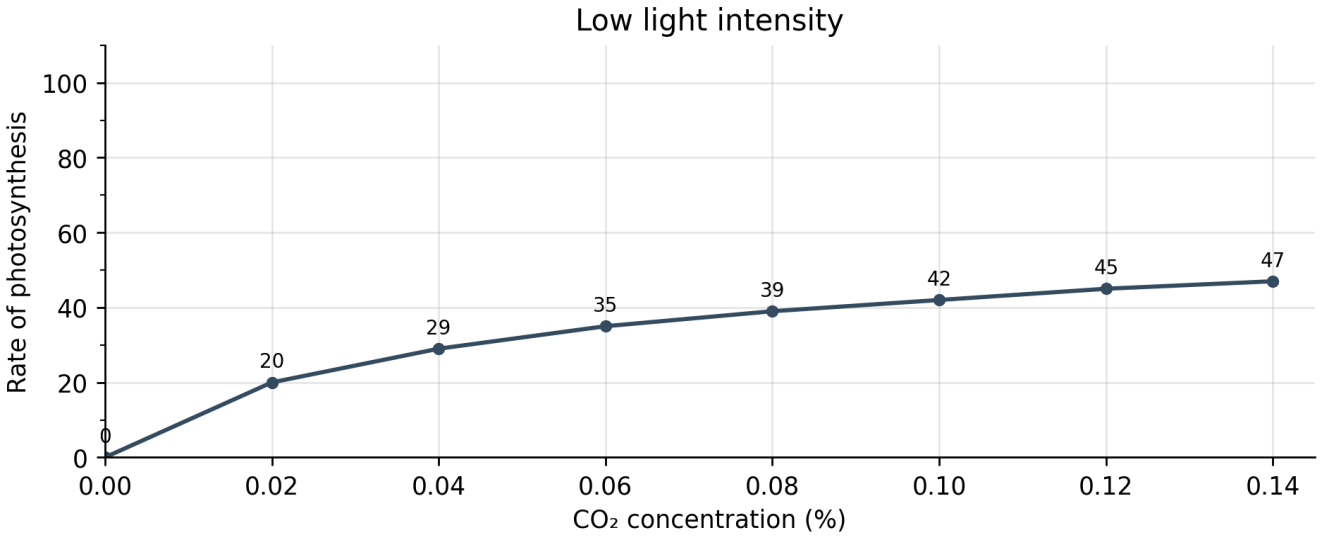
Phototropism: directional growth in response to light; for example, a shoot grows towards light.

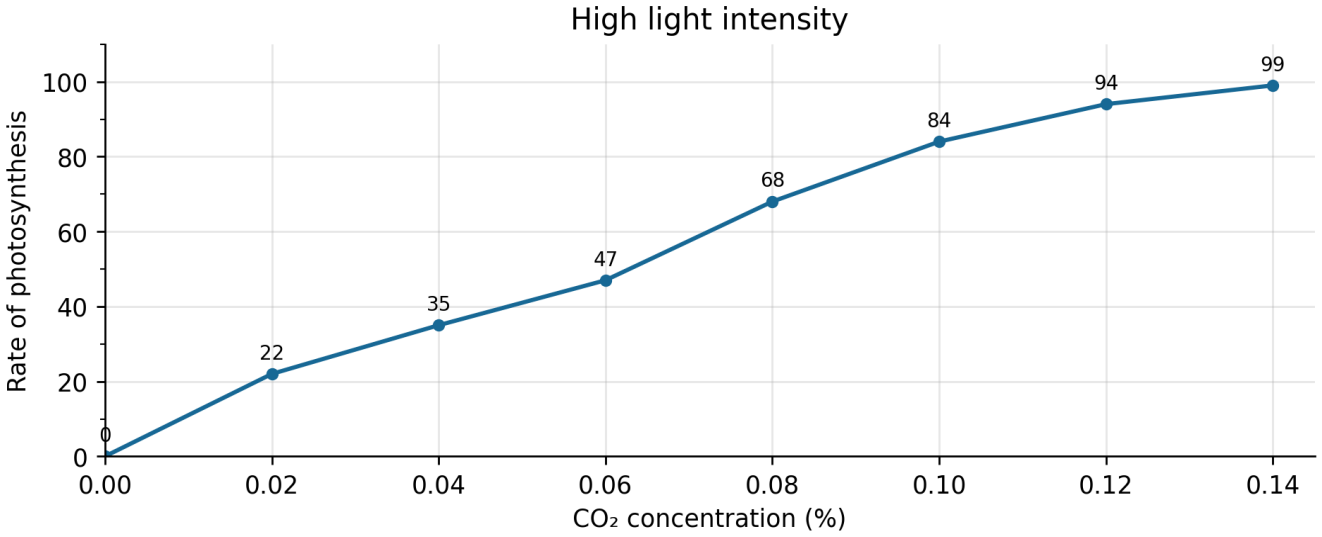
Other correct examples: hydrotropism (growth in response to a water/moisture gradient) or thigmotropism (growth in response to touch, as in coiling tendrils). Only one is required.

Memory aid: Positive = towards the stimulus. Negative = away from the stimulus. “Negative” does not mean harmful.

Q3(a). Two photosynthesis graphs and comparison [2]

CO_2 (%)	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14
Low light	0	20	29	35	39	42	45	47
High light	0	22	35	47	68	84	94	99



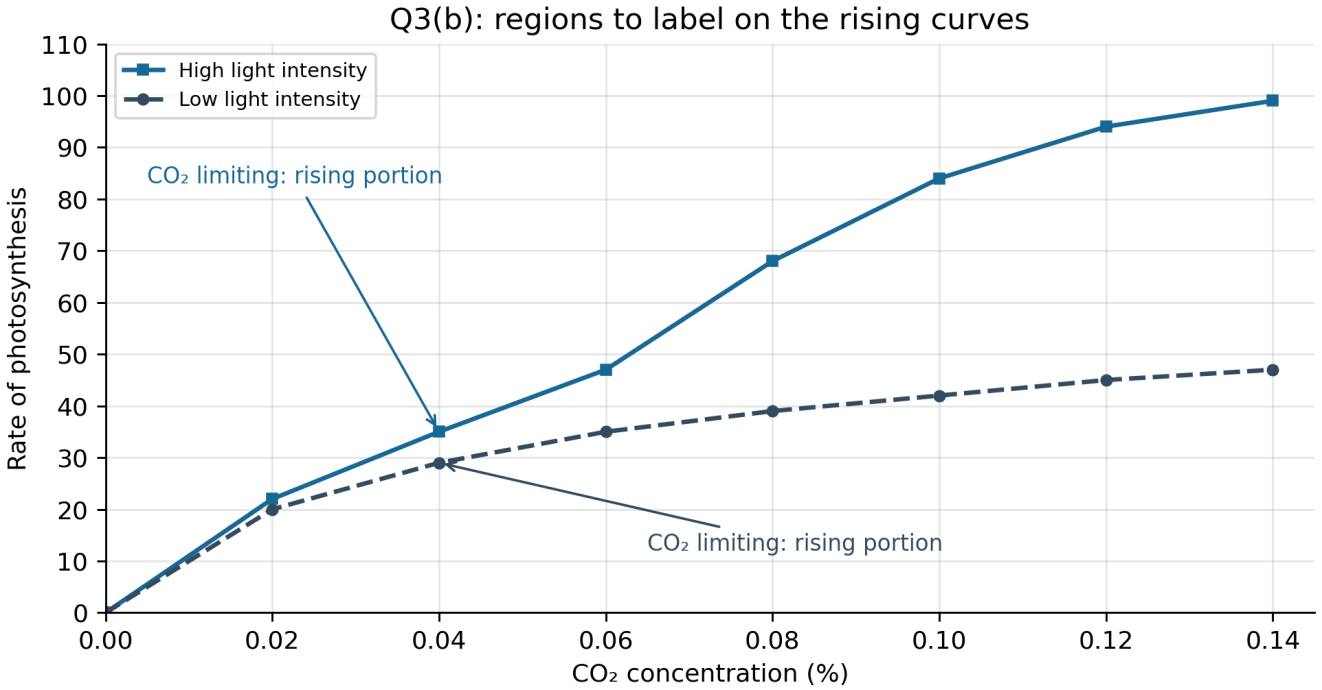


Comparison: At every non-zero CO_2 concentration shown, photosynthesis is faster in high light than in low light. For example, at 0.10% CO_2 the rates are **84** and **42**, respectively. At 0.00% both rates are zero.

Graph presentation: Both graphs use the same scales: x from 0.00% to 0.14%; y from 0 to 100. The table provides no rate units, so none have been invented. Every supplied value is plotted.

Q3(b-c). Limiting factors: labelled answer [4]

(b) Label the regions where CO_2 is limiting [2]



Write "CO_2 is a limiting factor" beside the upward-sloping part of each curve. The arrows above show clear locations for these labels. Increasing CO_2 increases the rate, showing that its availability restricts photosynthesis.

Clear example: Between 0.02% and 0.04% CO_2, the low-light rate rises from 20 to 29 and the high-light rate rises from 22 to 35. Point to these rising regions when explaining the answer.

Both curves still rise throughout the supplied range, up to 0.14%. The low-light curve becomes less steep at higher CO_2, indicating increasing restriction by light. The data do not locate a single exact cutoff at which CO_2 stops limiting, nor a perfectly horizontal plateau.

(c) Two other limiting factors and how a factor becomes limiting [2]

Two factors: temperature and **water availability**.

A limiting factor is a factor in shortest effective supply, or at an unfavourable level, that restricts the rate. Improving it increases the rate until another factor becomes limiting.

Temperature: Low temperature slows enzyme-controlled reactions. Increasing temperature towards the optimum can increase the rate; excessive temperature can inhibit photosynthesis and damage enzymes.

Water: Inadequate water can cause stomata to close, reducing CO_2 entry, and can restrict the plant's photosynthetic processes.

Short exam version: "Temperature and water supply may limit photosynthesis. A factor is limiting when it restricts the rate; improving that factor increases the rate until another factor becomes limiting."

Q4. Germination, organism and plant tissue [4]

(a) Identify the germination type and labels A, B and C [2]



Hypogeal germination. The cotyledons remain below the soil surface while the shoot grows above it. This visible position of the cotyledons is the key identifying feature.

Label in Figure 4.1	Answer	Role
A	Plumule / developing shoot	Forms the shoot system.
B	Cotyledon	Seed leaf containing or receiving food reserves.
C	Radicle / developing primary root	Forms the root system.

The source figure is small and pixelated. The labels above follow the visible pointers: A to the green shoot, B to the yellow cotyledon and C to the downward root.

(b) Identify the organism in Figure 4.2 [1]

Amoeba. Its irregular body bears **pseudopodia**, which help in movement and engulfing food. A nucleus is visible inside.

Plant tissue in Figure 4.3 [1]

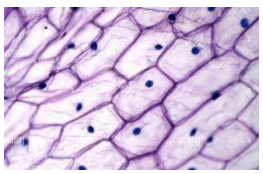
Epidermal tissue (onion epidermis). The micrograph shows a sheet of closely packed, brick-like plant cells with distinct cell walls and stained nuclei. The epidermis covers and protects the plant body.

Numbering note: The source prints an empty “c)” and then labels this tissue question “d)”. This is the one visible 1-mark tissue question; no missing question has been invented.

Identification checklist: Hypogeal = cotyledons below ground. Amoeba = pseudopodia. Epidermis = closely packed surface cells.



Amoeba: irregular outline and pseudopodia



Onion epidermis: closely packed surface cells

Q5(a). Diffusion of ink in water [2]



Figure: 5.1

Answer: Diffusion is faster in the glass containing **hotter water**. **Glass B appears to be the intended faster-spreading example** in Figure 5.1, where the ink forms a broader mixed cloud; in A it remains in more distinct narrow streams.

Reason: At higher temperature, particles have more average kinetic energy and move faster. The ink therefore spreads more rapidly from a region of higher concentration to regions of lower concentration.

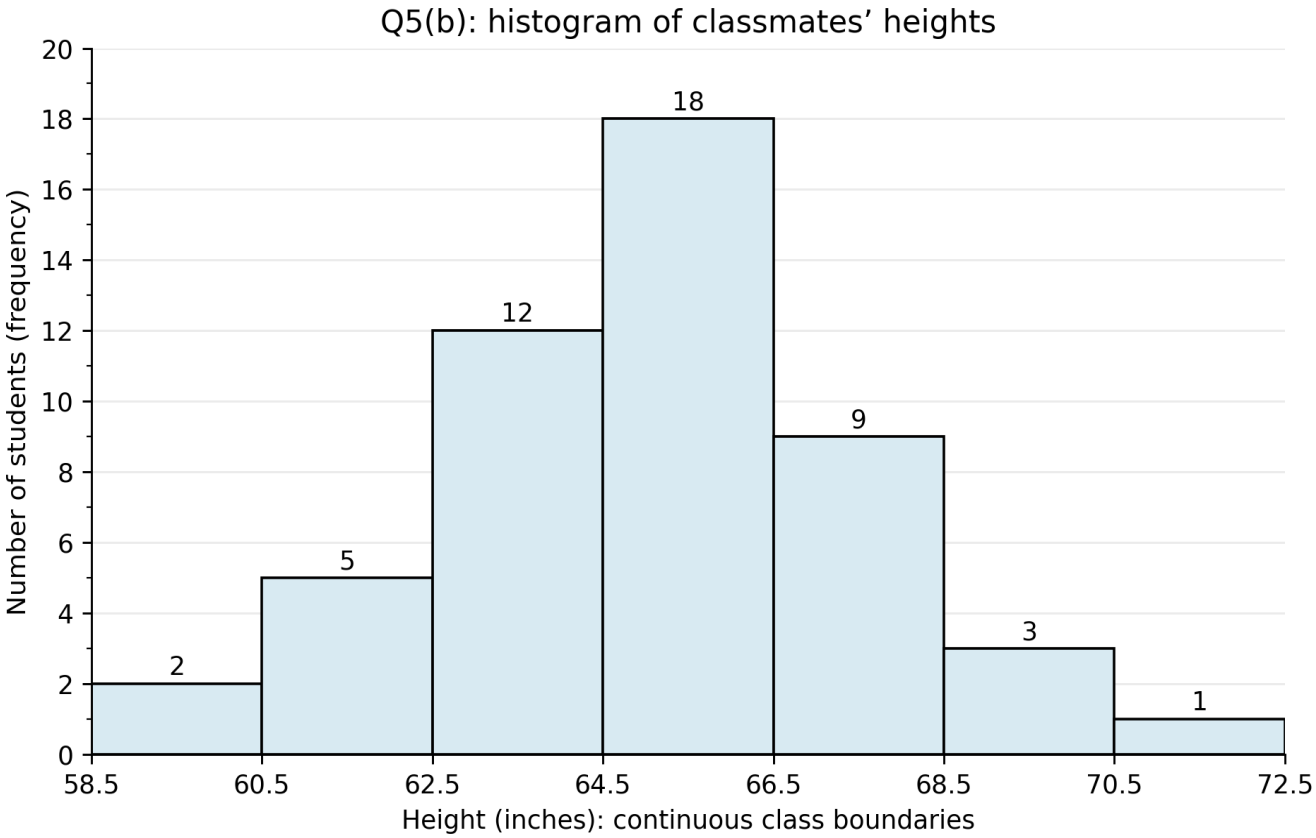
Figure limitation: The photograph has no temperature labels. The identification of B is inferred from the apparent spread in this still image; the secure scientific rule is **higher temperature → faster diffusion**, under comparable conditions. In an experiment, confirm the temperatures and compare at the same time.

Q5(b). Prepare the height data for a histogram [2]

Height range (inches)	Continuous boundaries (inches)	Frequency
59-60	58.5-60.5	2
61-62	60.5-62.5	5
63-64	62.5-64.5	12
65-66	64.5-66.5	18
67-68	66.5-68.5	9
69-70	68.5-70.5	3
71-72	70.5-72.5	1

For heights recorded to the nearest inch, subtract 0.5 from each lower class limit and add 0.5 to each upper limit. This creates adjoining intervals with no gaps. The finished histogram is on the next page.

Q5(b). Correctly drawn histogram [2]



How to reproduce the graph in the examination

1. Write the title **Histogram of classmates' heights**. Label the x-axis **Height (inches)** and the y-axis **Number of students**.

2. Mark the continuous boundaries 58.5, 60.5, 62.5, 64.5, 66.5, 68.5, 70.5 and 72.5 at equal distances on the x-axis.

3. Choose a suitable vertical scale, for example one major division = 2 students. Start the y-axis at zero.

4. Draw seven **touching bars**, each 2 inches wide. Their heights must be **2, 5, 12, 18, 9, 3 and 1**, respectively.

Numerical checks

Total number of students: 2 + 5 + 12 + 18 + 9 + 3 + 1 = **50**.

Modal class: 65-66 inches, containing 18 students. Its continuous interval is 64.5-66.5 inches.

Why the bars touch: Height is a continuous variable. Adjacent intervals meet at the class boundaries; this is a histogram, not a separated-bar chart.

All classes have equal widths, so frequencies are valid bar heights. For unequal class widths in another question, use frequency density = frequency ÷ class width.

Q6. Osmosis and the dicot seed [4]

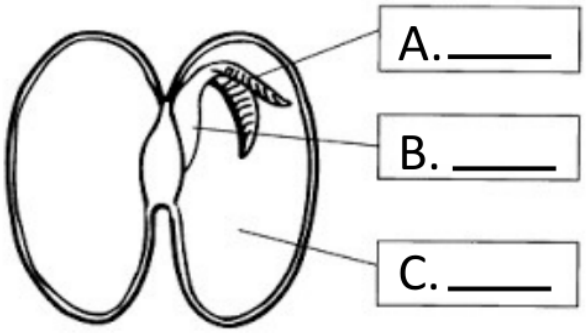
(a) Identify endosmosis and exosmosis, with reasons [2]

Beaker	Correct process	Reason and result
A: distilled water	Endosmosis	Water enters potato cells through their selectively permeable membranes, from higher water potential outside to lower water potential inside. Cells become turgid; the strip becomes firmer and generally gains mass.
B: concentrated salt solution	Exosmosis	Water leaves potato cells through their selectively permeable membranes, from higher water potential inside to lower water potential outside. Cells lose turgor; the strip becomes flaccid and generally loses mass.

Water direction: A: distilled water → potato cells. B: potato cells → salt solution.

Definition: Osmosis is the net movement of water across a selectively permeable membrane from higher to lower water potential.

(b) Identify seed parts A, B and C and give the role of C [2]



Label in Figure 6.2	Answer
A	Plumule
B	Radicle
C	Cotyledon

Role of C: The cotyledon stores food in a bean-type dicot seed. During germination, stored reserves are broken down and supplied to the growing embryo for respiration and growth until the young seedling can photosynthesise sufficiently.

Do not confuse the labels: In Q4, C is the root in a germinating seedling. In Q6, C is the large cotyledon inside a dissected dicot seed. Letters refer only to their own figure.

Final revision and source notes

Question	Key answers
Q1	Ethanol emulsion: fats; Biuret: proteins; Benedict's: reducing sugars; iodine: starch; amylase digests starch.
Q2	A: negative gravitropism; B: positive gravitropism; stimulus: gravity; another tropism: phototropism.
Q3	Plot both supplied series. Label rising portions as CO_2-limited. Other factors: temperature and water supply.
Q4	Hypogeal germination; A plumule, B cotyledon, C radicle; Amoeba; epidermal tissue.
Q5	Hotter water gives faster diffusion; B is the inferred intended glass. Histogram frequencies: 2, 5, 12, 18, 9, 3, 1. Total: 50.
Q6	A endosmosis; B exosmosis. Seed: A plumule, B radicle, C cotyledon. Cotyledon supplies stored food.

What makes an answer complete?

For an identification, give the exact name. For a reason, connect the cause to the observed result. For graphs, show a title, labels, uniform scales and correct plotting. For a histogram, use continuous class boundaries and touching bars.

Source and editorial notes

Question content, numerical data and original figures come from the uploaded model paper. The printed header states **Annual Examination 2027 & onwards**. The source contains an empty Q4(c), calls the tissue item Q4(d), and has minor figure/table cross-reference errors; all visible tasks have been answered once in sequence.

These are independently prepared model answers, not an official FBISE marking scheme. Figure 5.1's glass identification is an image-based inference because no temperatures are printed. The source's broad "sugar molecules" label does not justify saying amylase directly converts all starch into glucose.

Scientific references checked

• AQA: GCSE Biology practical handbook - food-test methods
• Oxford University Press Pakistan: practical food tests
• OpenStax Biology 2e: plant sensory responses and gravitropism
• University of Florida IFAS: hypogeous germination
• University of Iowa: diffusion in liquids at different temperatures

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

SSC-II BIOLOGY

PRACTICAL BASED ASSESSMENT

1st Annual Examination 2026 | Morning Group

Time allowed: 2:00 Hours **Total Marks: 20**

Sections A and B

Section A: 12 marks | Section B: 8 marks

SECTION – A (Marks 12)

Attempt all questions and write answers within spaces on E-Sheet. تمام سوالات کے جوابات اسی شیٹ پر دی گئی جگہوں میں لکھیں۔

Q.1 A student studied the effect of limiting factors on the rate of photosynthesis and obtained following data shown in **Table 1.1.**

ایک طالب علم نے محدود کنندہ عوامل کے اثر کا مطالعہ فوٹوسنتھیسس کی شرح پر کیا اور ذیل میں دیئے گئے اعداد و شمار حاصل کیئے، جیسا کہ جدول 1.1 میں دکھایا گیا ہے۔

Concentration of CO_2 (% age) کاربن ڈائی آکسائیڈ کا ارتکاز (%عمرہ)	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14
Rate of photosynthesis in low light intensity کم روشنی کی شدت میں فوٹوسنتھیسس کی شرح	0	20	29	35	39	42	45	47
Rate of photosynthesis in high light intensity زیادہ روشنی کی شدت میں فوٹوسنتھیسس کی شرح	0	22	35	47	68	84	94	99

Table 1.1

(a)	Draw graphs of the above data to show the rate of photosynthesis. Is the rate of photosynthesis the same in both graphs?	مندرجہ بالا اعداد و شمار کے گراف بنائیں تاکہ فوٹوسنتھیسس کی شرح کو ظاہر کیا جا سکے۔ کیا دونوں گرافوں میں فوٹوسنتھیسس کی شرح یکساں ہے؟
(b)	Which regions of the graphs show that carbon dioxide is a limiting factor?	گراف کے کون سے حصے ظاہر کرتے ہیں کہ کاربن ڈائی آکسائیڈ ایک محدود کنندہ عامل ہے؟
(c)	What is photosynthesis? How does a factor become a limiting factor for photosynthesis?	فوٹوسنتھیسس کیا ہے؟ کوئی عامل فوٹوسنتھیسس کے لیے محدود کنندہ عامل کیسے بن جاتا ہے؟

(2 × 3 = 6)

SECTION A (continued) **(6 marks)**

Q.2 A student studied the process of transpiration in plants using cobalt chloride paper as shown in Figure 2.1. **(2 × 3 = 6)**

ایک طالب علم نے پودوں میں ٹرانسپیریشن کے عمل کا مطالعہ کوبالٹ کلورائیڈ پیپر کے ذریعے کیا جیسا کہ شکل 2.1 میں دکھایا گیا ہے۔

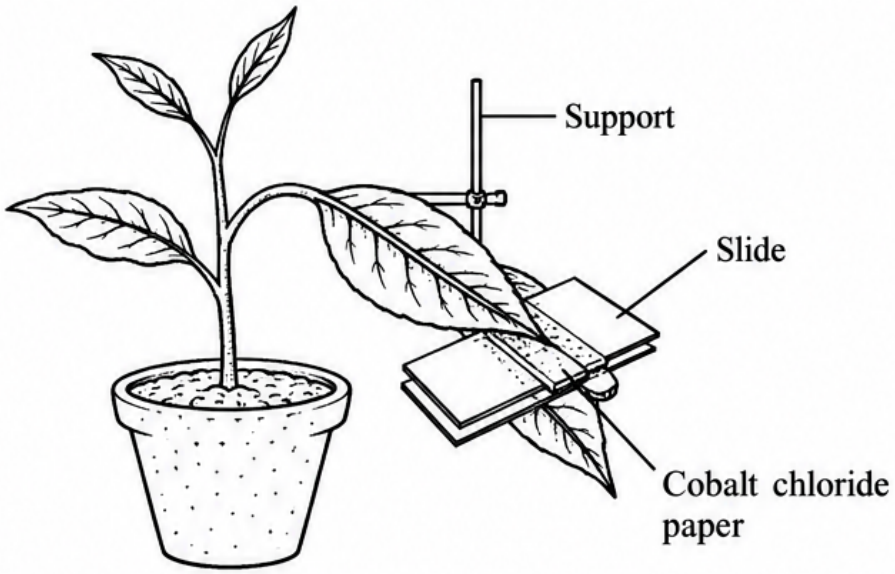


Figure 2.1

(a)	Write down a brief procedure of the experiment.	اس تجربے کا مختصر طریقہ کار لکھیں۔
(b)	Which surface of the epidermis changes the colour of cobalt chloride paper first and why?	ایپیڈرمس کی کون سی سطح کوبالٹ کلورائیڈ پیپر کا رنگ سب سے پہلے تبدیل کرتی ہے اور کیوں؟
(c)	Will similar results be obtained in summer and winter? Give a reason.	کیا گرمی اور سردی کے موسموں میں اس طرح کے مشابہ نتائج حاصل ہوں گے؟ وجہ بتائیں۔

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FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

SSC-II BIOLOGY — PRACTICAL BASED ASSESSMENT

1st Annual Examination 2026 | Morning Group

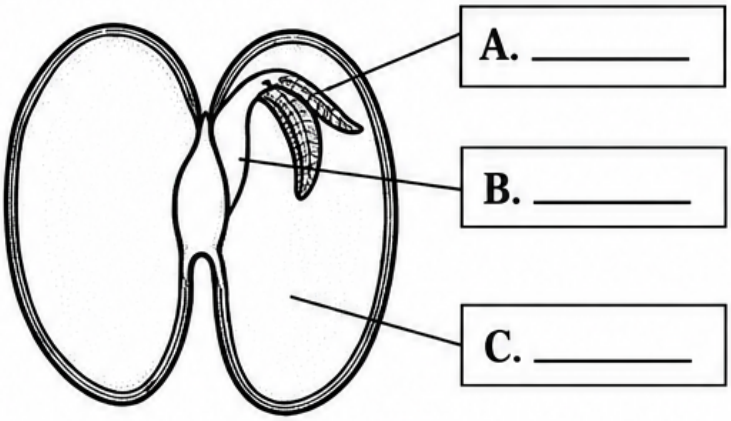
SECTION – B (Marks 8)

Q.3 **(2 + 1 + 1 = 4 marks)**

a. Identify the dissected seed as monocot or dicot and label three parts A, B and C indicated in diagram 3.1.

شکل 3.1 میں دکھائے گئے کٹے ہوئے بیج کی شناخت کریں کہ یہ ایک بیج پتی والا ہے یا دو بیج پتی والا ہے اور خاکے میں ظاہر کردہ تین حصوں A، B اور C کو لبیل کریں۔

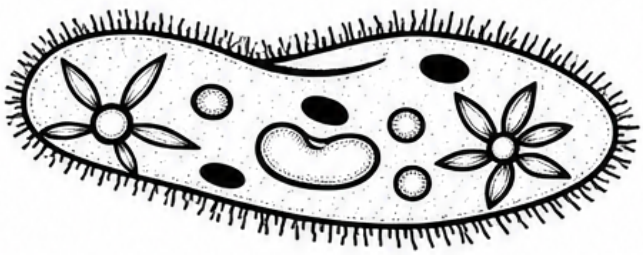
Diagram 3.1



b. Identify the organism shown in diagram 3.2.

شکل 3.2 میں دکھائے گئے جاندار کی شناخت کریں۔

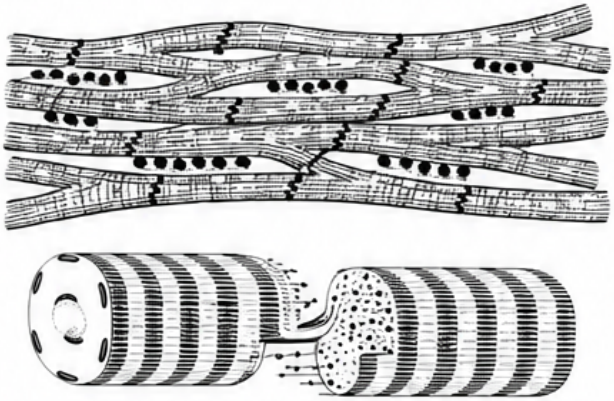
Diagram 3.2



c. Identify the type of animal tissue shown in diagram 3.3.

شکل 3.3 میں دکھائے گئے حیوانی بافت (نشو) کی قسم کی شناخت کریں۔

Diagram 3.3



SECTION B (continued) (2 + 2 = 4 marks)

Q.4

a. The figure 4.1 shows process of osmosis. Identify the glass beakers in which exosmosis and endosmosis will occur. Give reasons for your answer.

شکل 4.1 عمل اسموسس کو ظاہر کرتی ہے۔ ان شیشے کے برتن کی شناخت کریں جن میں خارج اسموسس (Exosmosis) اور داخل اسموسس (Endosmosis) وقوع پذیر ہوں گے۔ اپنے جواب کی وجوہات بیان کریں۔

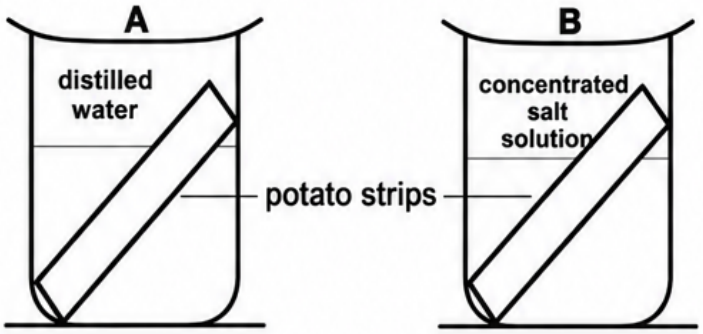


Figure 4.1

b. A student was studying continuous trait of weight of classmates and recorded data of weights of his classmates shown in table 4.2. Draw a histogram to represent data using suitable scale.

ایک طالب علم اپنے جماعت کے ساتھیوں کے وزن کے مسلسل صفت کا مطالعہ کر رہا تھا اور اس نے جدول 4.2 میں دکھائے گئے اپنے جماعت کے ساتھیوں کے وزن کا ڈیٹا ریکارڈ کیا۔ مناسب پیمانے (Scale) کا استعمال کرتے ہوئے ڈیٹا کی نمائندگی کے لیے ایک ہسٹوگرام بنائیں۔

Weight range (kg) وزن کی رینج	59-60	61-62	63-64	65-66	67-68	69-70	71-72
Number of students طلبہ کی تعداد	3	5	14	19	8	4	1

Table 4.2

Recreated from supplied paper images for educational use

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

SSC-II BIOLOGY — PRACTICAL BASED ASSESSMENT

1st Annual Examination 2026 | Evening Group

Time allowed: 2:00 Hours | **Section A: 12 marks | Section B: 8 marks** | **Total Marks: 20**

SECTION A (Marks 12)

Attempt all questions and write answers within provided spaces on E-Sheet. تمام سوالات کے جوابات ای شیٹ میں فراہم کردہ جگہ میں لکھیں۔

1. A student studied the rate of photosynthesis using hydrogen carbonate indicator / disc method and obtained the following data in **Table 1.1.**

ایک طالب علم نے بائیڈروجن کاربونیٹ انڈیکیٹر / ڈسک طریقہ استعمال کرتے ہوئے ضیائی تالیف کی شرح کا مطالعہ کیا اور درج ذیل اعداد و شمار جدول 1.1 میں حاصل کیے۔

Time (minutes) وقت (منٹ)	0	2	4	6	8	10	11	12
Number of floating discs فلوٹائی ڈسکس کی تعداد	0	1	3	5	7	8	10	12

Table 1.1

a. Make a graph of the above data to show rate of photosynthesis. Does the rate of photosynthesis remain same all the time according to graph?

مندرجہ بالا اعداد و شمار کو استعمال کرتے ہوئے ضیائی تالیف کی شرح کا گراف بتائیں۔ کیا گراف کے مطابق ضیائی تالیف کی شرح ہر وقت ایک جیسی رہتی ہے؟ **(2 × 3 = 6)**

b. Why do leaf discs rise in syringe during experiment? What is the purpose of bicarbonate indicator?

تجربے کے دوران سرنج میں پتے کے ڈسکس اوپر کیوں آتے ہیں؟ بائیکاربونیٹ انڈیکیٹر کا مقصد کیا ہے؟

c. How air pockets within leaves were removed to make leaf discs sink?

پتوں کے ڈسکس کو ڈوبانے کے لیئے ان کے اندر موجود ہوا کے خالی خانوں (ایئر پاکٹس) کو کس طرح نکالا گیا؟

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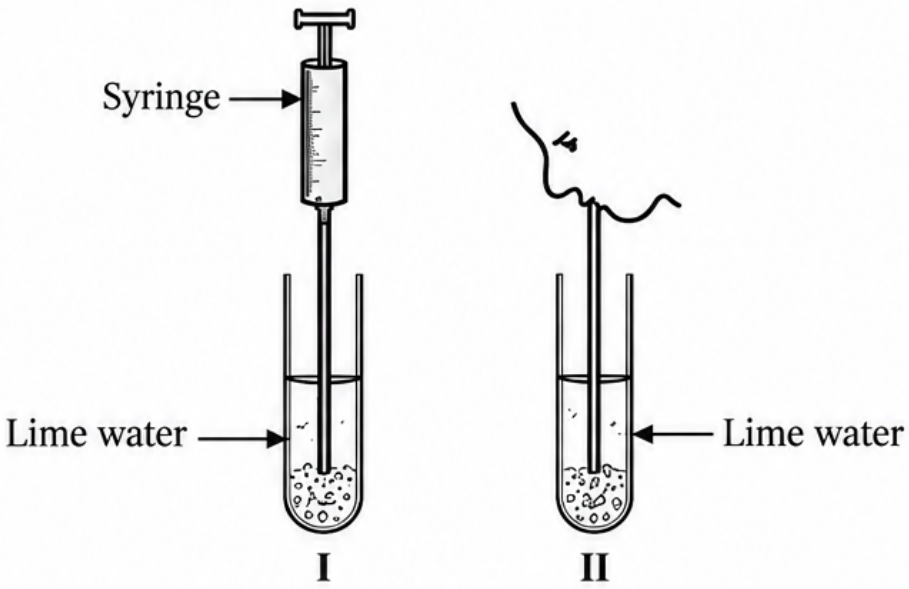
SECTION A (continued)

Q.2 **(2 × 3 = 6 marks)**

A student investigated that exhaled air contains carbon dioxide as shown in figure 2.1.

ایک طالب علم نے تحقیق کی کہ خارج کی گئی ہوا میں کاربن ڈائی آکسائیڈ موجود ہوتی ہے جس طرح شکل 2.1 میں دکھایا گیا ہے:

Figure 2.1



a.	Write down brief procedure of experiment?	تجربے کا مختصر طریقہ کار لکھیں۔
b.	What is percentage of carbon dioxide in exhaled air? Why exhaled air contains more carbon dioxide than inhaled air?	خارج کی گئی ہوا میں کاربن ڈائی آکسائیڈ کی شرح (فیصد) کتنی ہوتی ہے؟ اور خارج شدہ ہوا میں سانس کے ذریعے لی گئی ہوا کی نسبت زیادہ کاربن ڈائی آکسائیڈ کیوں ہوتی ہے؟
c.	In which flask lime water turns milky first i.e flask I or II and why?	کن فلاسک میں چونے کا پانی پہلے دودھیا ہوتا ہے، یعنی فلاسک I یا II؟ وجہ بیان کریں۔

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FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

SSC-II BIOLOGY — PRACTICAL BASED ASSESSMENT

1st Annual Examination 2026 | Evening Group

SECTION – B (Marks 8)

Q.3 **(2 + 1 + 1 = 4 marks)**

a. Identify the labelled parts P, Q, R and S in the dissected flower shown in Diagram 3.1.

شکل 3.1 میں کیے دکھائے گئے (ڈسیکنڈ) پھول میں P، Q، R اور S سے موسوم حصوں کی شناخت کریں۔

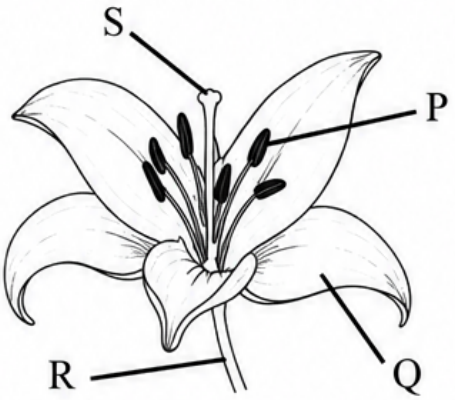


Diagram 3.1

b. Identify the organism shown in diagram 3.2.

شکل 3.2 میں دکھائے گئے جاندار کی شناخت کریں۔

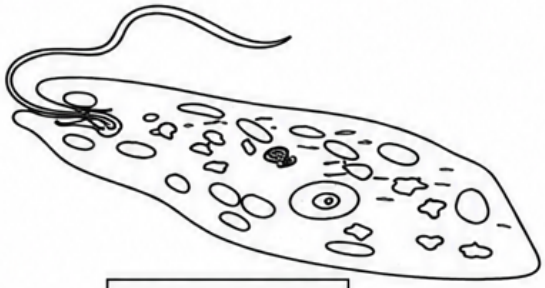


Diagram 3.2

c. Identify the type of animal tissue shown in diagram 3.3.

شکل 3.3 میں دکھائی گئی حیوانی بافت (تشو) کی قسم کی شناخت کریں۔

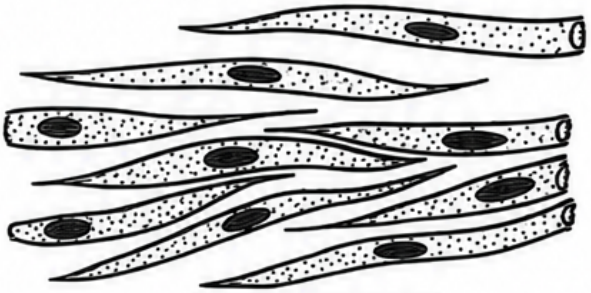


Diagram 3.3

Recreated from supplied paper images for educational use

Q4 (2 + 2 = 4 marks)

(a) Figure 4.1 shows diffusion of dye in two glass beakers of water at different temperatures. Identify the glass beaker with faster diffusion?
Give reason of your answer.

(الف) شکل 4.1 میں مختلف درجہ حرارت پر پانی سے بھرے دو شیشے کے بیکرز میں رنگ کے پھیلاؤ (انتشار) کو دکھایا گیا ہے۔ زیادہ تیزی سے انتشار والا شیشے کا بیکر کون سا ہے؟ اپنے جواب کی وجہ بیان کریں۔

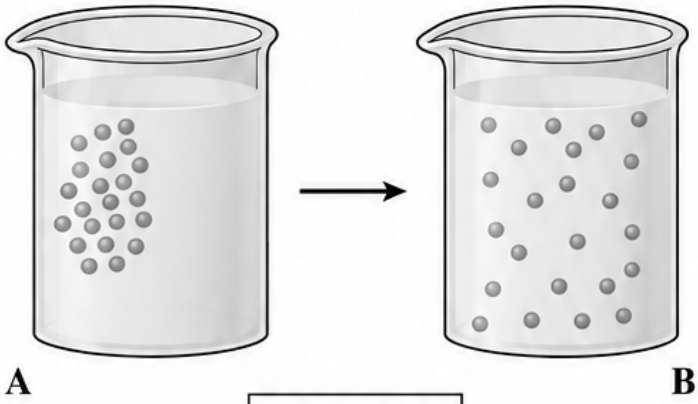


Figure 4.1

(b) A student was studying continuous trait of height of classmates and recorded data of heights of his classmates in table 4.2. Draw a histogram to represent data selecting suitable scale.

(ب) ایک طالب علم اپنے ہم جماعتوں کے قد (مسلسل صفت) کا مطالعہ کر رہا تھا اور اس نے اپنے ہم جماعتوں کے قد کا ڈیٹا جدول 4.2 میں درج کیا۔ مناسب اسکیل منتخب کرتے ہوئے اس ڈیٹا کی نمائندگی کے لیے ایک ہسٹوگرام رسم کریں۔

Height range (in) قدوں کی رینج	59-60	61-62	63-64	65-66	67-68	69-70	71-72
Number of students طلباء کی تعداد	3	5	14	19	8	4	1

Table 4.2

FBISE SSC-II Biology

Practical Based Assessment Complete worked solutions

1st Annual Examination 2026
Morning Group followed by Evening Group

Paper	Section A	Section B	Total
Morning Group	Q1 + Q2 = 12	Q3 + Q4 = 8	20 marks
Evening Group	Q1 + Q2 = 12	Q3 + Q4 = 8	20 marks

Scope and order

These solutions cover every visible question and subpart in the supplied Morning and Evening papers. Both supplied papers were identified as **1st Annual 2026**. A separate 2nd Annual paper was not supplied; the Evening paper is therefore kept under its correct title.

How to use these answers

• Write the direct answer first, then the observation and scientific reason. Key terms are bold for revision.

• Copy graphs with a ruler, labelled axes, a uniform scale and accurately plotted points. Histograms need touching bars.

• For identification questions, follow the pointer to its endpoint. Add a distinguishing feature if space permits.

• The answers are exam-focused model solutions, not an official FBISE marking scheme. Marks depend on the actual question and examiner’s scheme.

Page guide

Pages	Content
2-5	Morning: photosynthesis; transpiration; identification; osmosis; histogram
6-10	Evening: floating discs; CO_2 test; identification; diffusion; histogram
11	Revision checklist, scientific clarifications and references

Prepared for Fakhar STEM Sphere • Simple English with complete scientific reasoning.

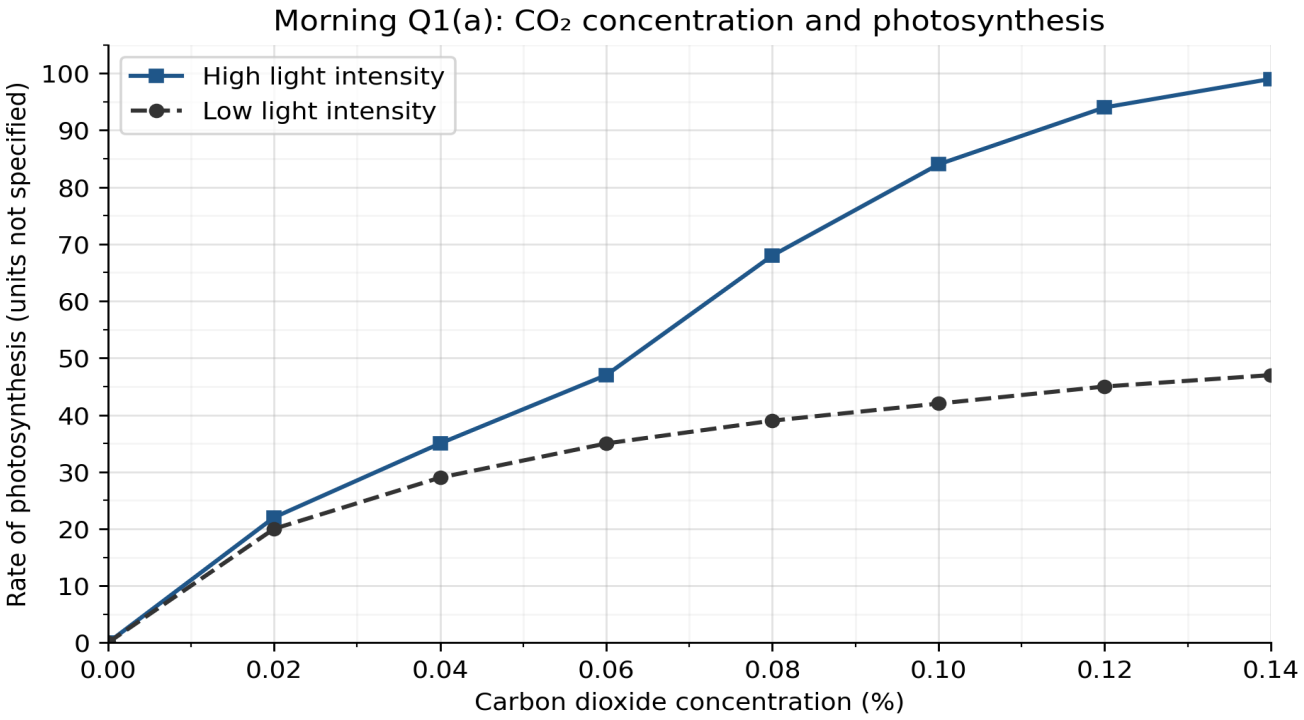
MORNING GROUP • SECTION A • Q1 (6 marks)

Q1. CO_2 concentration and photosynthesis

Question: Use Table 1.1 to draw graphs, compare the rates, identify CO_2-limited regions and explain photosynthesis and limiting factors.

CO_2 (%)	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14
Low light	0	20	29	35	39	42	45	47
High light	0	22	35	47	68	84	94	99

(a) Graph and comparison



No, the two rates are not the same. At every non-zero CO_2 concentration shown, the rate is higher in high light than in low light. At 0.00% CO_2, both rates are zero. For example, at 0.10% CO_2, the rates are **84** and **42**, respectively.

Graph method: Put CO_2 concentration on the horizontal axis and rate on the vertical axis. Use x = 0.00 to 0.14% and y = 0 to 100. Plot both series with different symbols and a key. Points are joined here to preserve the measured data; do not invent units for the rate.

(b) Regions where CO_2 is a limiting factor

CO_2 is limiting in the **rising portions**, because increasing CO_2 increases the rate. This is especially clear at low concentrations. Both curves continue to rise through the supplied range, although the low-light curve becomes much less steep at higher CO_2. Thus light becomes increasingly restrictive there; the data do not show a perfectly horizontal plateau.

MORNING GROUP • SECTION A

Q1(c) and Q2. Transpiration

Q1(c). What is photosynthesis? How does a factor become limiting?

Photosynthesis is the process by which green plants use light energy, absorbed by chlorophyll, to make glucose from carbon dioxide and water, releasing oxygen.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
Conditions: light energy and chlorophyll. This is the standard school-level net equation.

A **limiting factor** is the factor in shortest effective supply that restricts the rate of a process. Increasing that factor increases the rate until another factor becomes limiting. For example, adding CO_2 can increase photosynthesis, but in dim light the rate eventually becomes restricted mainly by the light supply.

Q2(a). Brief procedure of the cobalt chloride experiment (2 marks)

1. Take a healthy, well-watered potted plant. Choose a dry, intact leaf and use forceps to handle two equal pieces of **dry blue cobalt chloride paper**.

2. Place one piece against the upper surface and the other against the lower surface of the same leaf. Cover each paper with a dry glass slide and secure gently with clips, without damaging the leaf.

3. Start timing together and note the time each paper takes to turn from **blue to pink**. Compare the two times under the same conditions.

Observation and conclusion: The lower-surface paper usually turns pink first in a typical dorsiventral land-plant leaf. The colour change detects water lost by transpiration; a shorter time indicates faster water loss.

Q2(b). Which surface changes the paper first, and why? (2 marks)

The paper against the **lower epidermis** usually changes from blue to pink first. In a typical dorsiventral leaf, the lower epidermis has more stomata, so more water vapour escapes there. The vapour moistens the cobalt chloride paper and turns it pink.

Concept: Transpiration is loss of water vapour from aerial plant parts, mainly through stomata. The paper detects **water**, not CO_2.

Q2(c). Will summer and winter give the same results? (2 marks)

Usually not. Under typical warmer, drier summer conditions, evaporation and transpiration are faster, so the paper becomes pink sooner. Under cooler winter conditions, the change generally takes longer. The lower-versus-upper pattern normally remains the same for the same type of leaf.

The comparison assumes similar light, wind, leaf condition and water supply. Humidity and stomatal closure can change the result, so “summer always gives more transpiration” is too absolute.

MORNING GROUP • SECTION B

Q3. Identification and Q4(a). Osmosis

Q3(a). Identify the dissected seed and label A, B and C (2 marks)

Identification: dicotyledonous seed (dicot), such as bean. Two large cotyledons are visible.

Label in supplied diagram	Answer	Function
A	Plumule	Embryonic shoot; develops into the shoot system.
B	Radicle	Embryonic root; develops into the primary root.
C	Cotyledon	Seed leaf; stores food in a bean seed and nourishes the embryo.

Label B follows the lower embryonic axis in the supplied drawing; it does not indicate the large food-storing cotyledon.

Q3(b). Identify the organism in Diagram 3.2 (1 mark)

Paramecium. It has a slipper-shaped body covered with short **cilia**. Cilia help in movement and directing food towards the oral groove. Star-shaped contractile vacuoles are also characteristic.

Q3(c). Identify the animal tissue in Diagram 3.3 (1 mark)

Cardiac muscle tissue. Identify it by branching, striated fibres joined at **intercalated discs**. It forms the muscular wall of the heart and contracts involuntarily to pump blood. Striations alone do not distinguish it from skeletal muscle; branching and intercalated discs are important.

Q4(a). Identify endosmosis and exosmosis in the potato experiment (2 marks)

Beaker	Process	Reason and expected observation
A: distilled water	Endosmosis	Water enters potato cells by osmosis. The outside has higher water potential than the cell sap. Cells become turgid; the strip becomes firmer and generally gains mass.
B: concentrated salt solution	Exosmosis	Water leaves potato cells by osmosis. The outside has lower water potential than the cell sap. Cells lose turgor; the strip becomes soft/flaccid and generally loses mass.

Definition: Osmosis is the net movement of water through a selectively permeable membrane from higher water potential to lower water potential.

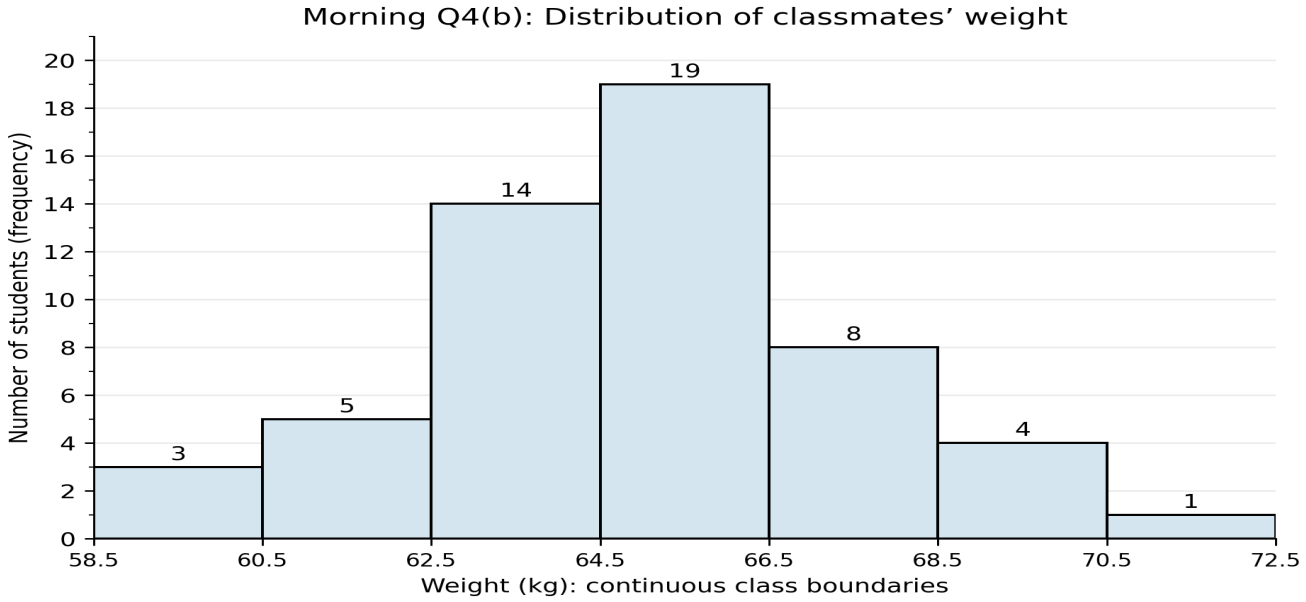
Write the direction explicitly: A: water outside → potato cells. B: potato cells → outside salt solution. The membrane is selectively permeable; the potato strip itself is not the membrane.

MORNING GROUP • SECTION B • Q4(b) (2 marks)

Histogram of classmates' weights

Question: Draw a histogram for the weight data in Table 4.2, using a suitable scale.

Weight range (kg)	Continuous boundaries (kg)	Frequency
59-60	58.5-60.5	3
61-62	60.5-62.5	5
63-64	62.5-64.5	14
65-66	64.5-66.5	19
67-68	66.5-68.5	8
69-70	68.5-70.5	4
71-72	70.5-72.5	1



Steps that secure the graph marks

1. Treat the stated integer classes as measurements recorded to the nearest kilogram. Subtract 0.5 from each lower limit and add 0.5 to each upper limit to make continuous boundaries.

2. Label the horizontal axis **Weight (kg)** and vertical axis **Number of students**. A convenient scale is one major class width = 2 kg and one major vertical division = 2 students.

3. Draw seven **touching bars** with heights 3, 5, 14, 19, 8, 4 and 1. All classes have equal width, so frequency heights are appropriate.

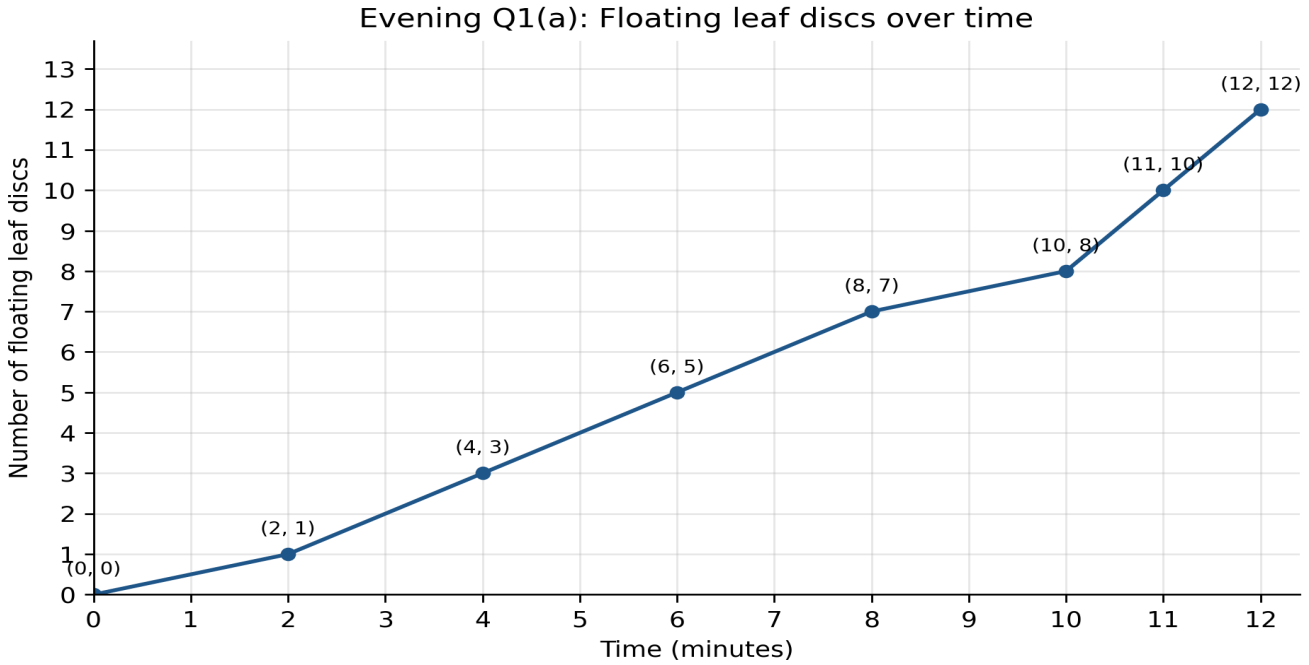
Check: Total students = 54. The modal class is 65-66 kg, with 19 students. The x-axis starts at 58.5 kg; zero is not a data class.

EVENING GROUP • SECTION A • Q1 (6 marks)

Q1(a). Floating leaf discs: correct graph

Question: Graph the supplied floating-disc data and state whether the rate remains the same throughout.

Time (min)	0	2	4	6	8	10	11	12
Floating discs	0	1	3	5	7	8	10	12



Exam-ready conclusion

No, the inferred rate does not remain constant. The graph has different slopes over different intervals. For example, from 0 to 2 minutes the count rises from 0 to 1, whereas from 10 to 11 minutes it rises from 8 to 10. A steeper segment represents a faster increase in the number of discs floating.

Example comparison:
0-2 min: $(1 - 0) \div (2 - 0) =$ **0.5 disc min^{-1}**.
10-11 min: $(10 - 8) \div (11 - 10) =$ **2 discs min^{-1}**.

Graph accuracy

• Use **actual time spacing**: 10, 11 and 12 minutes are one minute apart, while earlier readings are two minutes apart. Do not space every reading equally.

• The y-axis is **number of floating discs**, not volume of oxygen or a directly measured photosynthetic rate.

• Floating-disc counts are an indirect indicator of photosynthetic activity. The interval calculations describe changes in floating counts; they are not direct oxygen-production measurements.

EVENING GROUP • SECTION A • Q1 continued

Q1(b-c). Why discs float and how air is removed

(b)(i). Why do leaf discs rise during the experiment?

In light, the discs carry out **photosynthesis** and produce **oxygen**. Oxygen accumulates in the internal air spaces, increasing buoyancy and lowering the average density of the discs. When sufficiently buoyant, they rise to the surface.

Short exam answer: Oxygen produced during photosynthesis collects inside the discs and makes them buoyant, so they float. It is not CO_2 released by photosynthesis.

(b)(ii). What is the purpose of bicarbonate indicator?

A true **hydrogencarbonate (bicarbonate) indicator detects changes in CO_2 concentration** by a colour change. It is red/orange near atmospheric CO_2, yellow when CO_2 rises and purple when CO_2 falls. CO_2 removal by photosynthesis can therefore be detected by a shift towards purple.

Reagent	Correct purpose
Hydrogencarbonate indicator	Detects changes in CO_2 by colour change.
Sodium hydrogencarbonate solution in the floating-disc method	Provides a source of dissolved CO_2 for photosynthesis. It is not, by itself, a colour indicator.

Why both are explained: The supplied paper combines the words “hydrogen carbonate indicator / disc method” with floating-disc data. If asked specifically about the **indicator**, state CO_2 detection. If the reagent is sodium bicarbonate **solution** in the disc experiment, state CO_2 supply. This distinction prevents an incorrect answer caused by the wording.

(c). How are air pockets removed to make the discs sink? (2 marks)

1. Put equal-sized leaf discs and sodium hydrogencarbonate solution in a syringe. Expel unwanted air from the barrel.

2. Close the syringe tip with a finger and pull back the plunger to create **reduced pressure / a partial vacuum**. Hold briefly and gently swirl. Air leaves the leaf spaces as bubbles.

3. Release the vacuum so solution enters the spaces that contained air. Repeat if necessary. The infiltrated discs become denser and **sink**.

Useful controls: Keep leaf species, disc size, bicarbonate concentration, solution volume, temperature and light distance constant when comparing treatments. A dark control normally produces little or no photosynthetic oxygen.

EVENING GROUP • SECTION A • Q2 (6 marks)

Q2. CO_2 in exhaled air: limewater test

(a). Brief procedure (2 marks)

1. Put equal volumes of fresh, clear limewater of the same concentration into two clean test tubes, I and II.

2. Use a syringe to bubble ordinary atmospheric air through tube I. Pass exhaled air through a delivery tube into tube II, as shown in the paper.

3. Compare the time, or comparable amount of gas, needed for each sample of limewater to turn milky. Keep other conditions the same.

Observation: The limewater receiving exhaled air becomes milky sooner. This shows that exhaled air contains more CO_2 than atmospheric air.

(b). CO_2 percentage and why it is higher (2 marks)

Exhaled air contains approximately **4% CO_2**, compared with about **0.04%** in inhaled atmospheric air. These are approximate school-level values, not fixed values for every breath.

Body cells produce CO_2 during **aerobic cellular respiration**. The blood carries this CO_2 to the lungs; it diffuses into the alveoli and is breathed out. Therefore, exhaled air has more CO_2 than inhaled air.

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy

(c). Which flask turns milky first, and why? (2 marks)

Flask / tube II, receiving exhaled air, turns milky first. Its gas has a higher CO_2 concentration. CO_2 reacts with calcium hydroxide in limewater to form an insoluble white precipitate of **calcium carbonate**, causing the milky appearance.

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$

Apparatus in supplied figure	Gas	Expected result
I: syringe side	Atmospheric air	Milky more slowly, under comparable conditions.
II: mouth / exhaled-air side	Exhaled air	Milky first because more CO_2 is supplied.

Identification rule: Follow the gas source if labels differ in a future paper: the **exhaled-air** tube changes first, regardless of whether it is called I, II, A or B.

Practical note: Blow gently through the delivery tube; do not suck limewater into the mouth. Stop once the milky endpoint is observed.

EVENING GROUP • SECTION B

Q3. Identification and Q4(a). Diffusion

Q3(a). Label P, Q, R and S on the flower (2 marks)

Label	Answer for the supplied pointer	Function
P	Anther	Produces pollen grains.
Q	Petal (broad perianth segment shown)	Helps attract pollinators in a typical insect-pollinated flower.
R	Pedicel / flower stalk	Supports and attaches the flower to the plant.
S	Stigma	Receives pollen grains during pollination.

Pointer check: P ends at the dark pollen-bearing tip, not its thin supporting filament. S ends at the top of the central female organ, not the ovary. Q points to the broad petal-like floral segment in the supplied drawing.

Q3(b). Identify the organism in Diagram 3.2 (1 mark)

Euglena. The drawing shows an elongated, spindle-like unicellular organism with a long **flagellum**. Euglena has chloroplasts and an eyespot; the flagellum helps it move. Do not identify it as Paramecium, which has many short cilia.

Q3(c). Identify the tissue in Diagram 3.3 (1 mark)

Smooth muscle tissue / unstriated muscle. Its cells are spindle-shaped with tapering ends, each normally having a single central nucleus and no visible cross-striations. It works involuntarily, for example in the intestinal wall to move food.

Q4(a). Which beaker has faster dye diffusion, and why? (2 marks)

Beaker B. In B the dye particles are more widely distributed through the water, indicating faster spreading for a comparable observation time. In A they remain concentrated in a smaller region.

At a **higher temperature**, particles have more average kinetic energy and move more rapidly, so diffusion is faster. Therefore B represents the faster-diffusion condition, consistent with warmer water in this comparison.

Definition: Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration, down the concentration gradient.

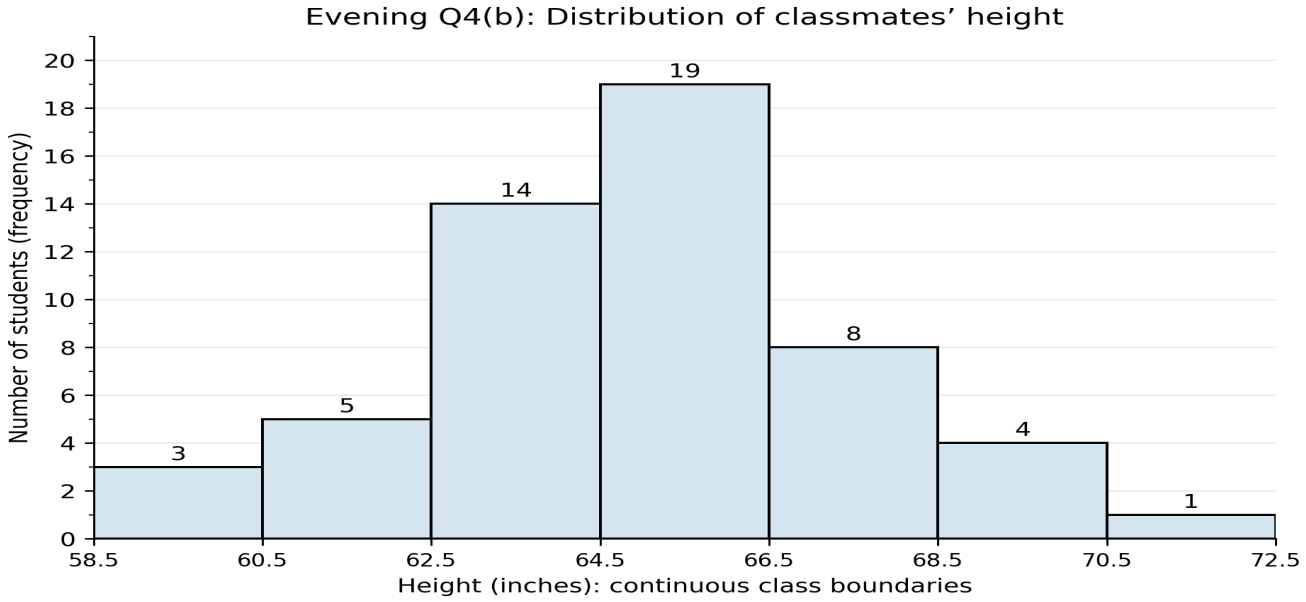
Do not invent temperatures: The supplied diagram does not give numerical temperatures. Write “higher temperature / warmer water”, not an unsupported number. The comparison assumes the beakers were observed after the same time with other conditions controlled.

EVENING GROUP • SECTION B • Q4(b) (2 marks)

Histogram of classmates' heights

Question: Draw a histogram for the height data in Table 4.2, using a suitable scale.

Height range (inches)	Continuous boundaries (inches)	Frequency
59-60	58.5-60.5	3
61-62	60.5-62.5	5
63-64	62.5-64.5	14
65-66	64.5-66.5	19
67-68	66.5-68.5	8
69-70	68.5-70.5	4
71-72	70.5-72.5	1



Construction and checks

1. For heights recorded to the nearest inch, convert the limits to continuous boundaries by subtracting or adding 0.5 inch.

2. Label x as **Height (inches)** and y as **Number of students**. Draw each bar over the stated boundary interval, with no gaps.

3. Use frequencies 3, 5, 14, 19, 8, 4 and 1. The maximum height is 19 students for the 64.5-66.5-inch bar.

Check: Total = 54 students; modal class = 65-66 inches. The shape matches the Morning histogram because the numerical frequencies match, but the measured variable and unit are different.

Histogram rule: Equal class widths permit frequency on the vertical axis. If a future paper has unequal widths, use frequency density = frequency ÷ class width, so bar areas represent frequencies.

REVISION AND VERIFICATION

Final checks before the PBA

Topic	Essential point
Photosynthesis	CO_2 + water → glucose + oxygen, using light and chlorophyll.
Limiting factor	Increasing the limiting factor increases rate until another factor restricts it.
Cobalt chloride paper	Dry blue → moist pink; usually lower leaf surface changes sooner.
Osmosis	Water moves through a selectively permeable membrane down water potential.
Floating discs	Vacuum infiltration makes discs sink; photosynthetic oxygen makes them float.
Limewater	CO_2 produces insoluble calcium carbonate: clear → milky.
Morning identification	Dicot; A plumule, B radicle, C cotyledon; Paramecium; cardiac muscle.
Evening identification	P anther, Q petal, R pedicel, S stigma; Euglena; smooth muscle.
Diffusion	Higher temperature generally increases diffusion rate.
Histograms	Continuous boundaries; touching bars; correct variable and units.

Important scientific clarifications

The supplied photosynthesis-rate table gives no rate units, so none have been invented. The Evening floating-disc graph uses real time spacing. CO_2 limitation is inferred from rising graph regions, not from an invented flat plateau. The two bicarbonate reagents are distinguished on page 7. Identification follows the original supplied drawings.

Sources checked for the scientific explanations

Question content and numerical data: the user-supplied Morning and Evening paper images. The solution is independently prepared; an official FBISE answer key was not supplied.

• Cornell Institute for Biology Teachers: floating-leaf-disc assay
• Science & Plants for Schools: photosynthesis and leaf adaptation
• Cambridge International: hydrogencarbonate indicator colour changes
• US National Cancer Institute, SEER: distinguishing muscle tissues
• Royal Society of Chemistry: CO_2, water and limewater
• Oxford University Press Pakistan: transpiration practical